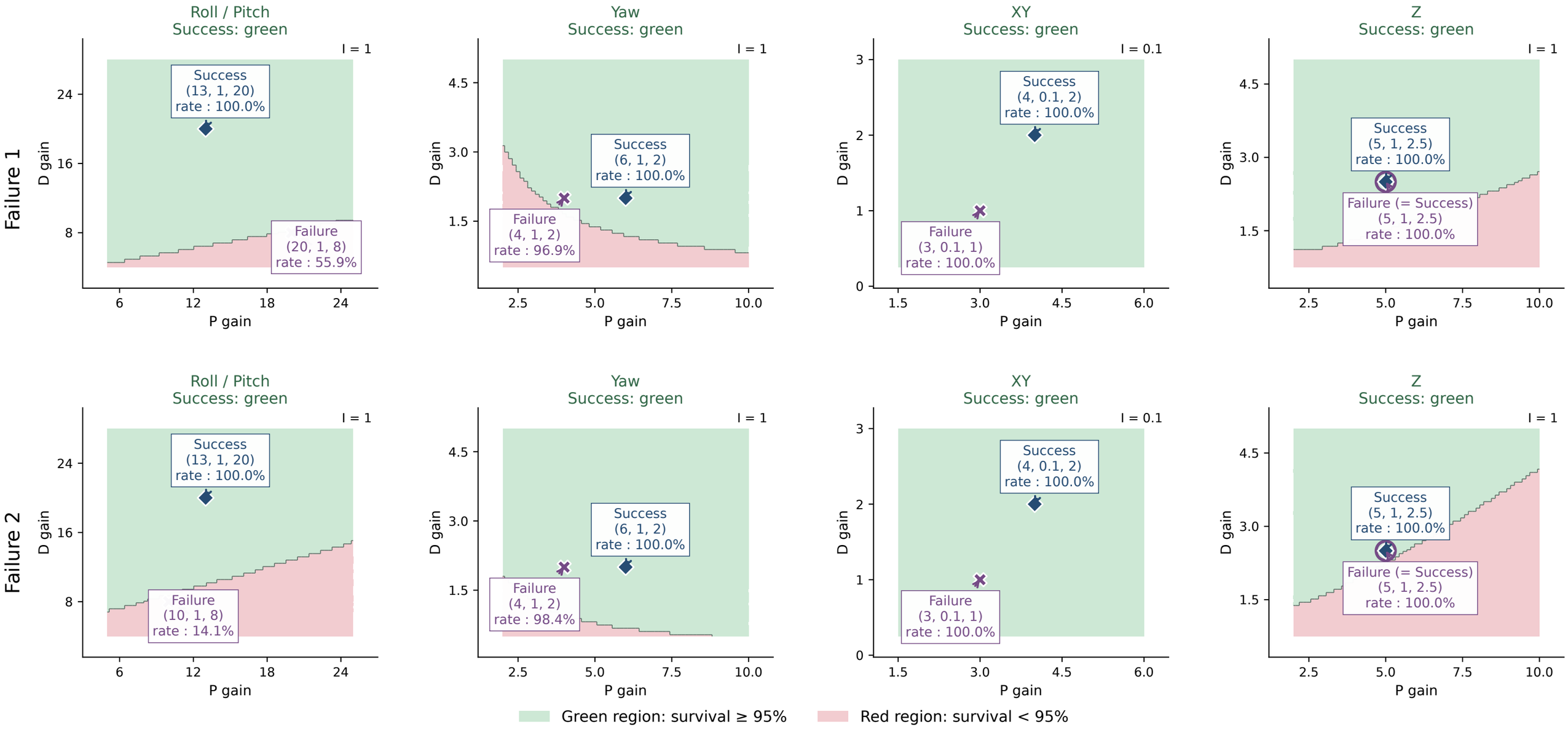
All PID gains: Success is in the green region on every axis
Failure 1
Failure 2
Roll / Pitch
Success: green
Yaw
Success: green
XY
Success: green
Z
Success: green
I = 1
I = 0.1
D gain
P gain
Success
(13, 1, 20)
rate : 100.0%
Failure
(20, 1, 8)
rate : 55.9%
Success
(6, 1, 2)
rate : 100.0%
Failure
(4, 1, 2)
rate : 96.9%
Success
(4, 0.1, 2)
rate : 100.0%
Failure
(3, 0.1, 1)
rate : 100.0%
Success
(5, 1, 2.5)
rate : 100.0%
Failure (= Success)
(5, 1, 2.5)
rate : 100.0%
Failure
(10, 1, 8)
rate : 14.1%
Failure
(4, 1, 2)
rate : 98.4%
Green region: survival ≥ 95%
Red region: survival < 95%
Each exact P–D slice passes through both reference points. Purple marks identify the failure-flight gains; Z gains coincide.
Each case uses its own posterior. Other controller gains are fixed at the successful-flight values.